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Research Interests

Climate sensitivity, cloud feedback, radiative forcing

Education

Ph.D., Atmospheric Sciences, University of Washington, Dec 2010

M.S., Atmospheric Sciences, University of Washington, Dec 2007

B.S., Meteorology, Pennsylvania State University, May 2004

Professional Experience

Research scientist, Lawrence Livermore National Laboratory, Apr 2013 - present.

Postdoctoral research scholar, Lawrence Livermore National Laboratory, Jan 2011 - Mar 2013.

Graduate research assistant, Dept. of Atmospheric Sciences, Univ. of Washington, Sep 2004 - Dec 2010.

Submitted Publications

Asterisks indicate current or former advisees

Norris, J, S. Rahimi , L. Huang, C. Thackeray, B. Bass , **M D. Zelinka**, and A. Hall, 2026: High vulnerability of Western United States mountain ranges to heavy rainfall events under warming, submitted.

Van Loon, S., M. Rugenstein, **M. D. Zelinka**, and T. Andrews, 2026: Recent Weakening of the Global Radiative Feedback, *Geophys. Res. Lett.*, submitted.

Covey, C., J. Lee, D. D. Lucas, K. E. Taylor, R. Klein, and **M. D. Zelinka**, 2026: Comparing the Arrhenius (1896) Model of Greenhouse Warming with Modern Climate Models, submitted.

Ceppi, P., A. Bodas-Salcedo, **M. D. Zelinka**, T. Andrews, F. Brient, R. Chadwick, J. M. Gregory, Y.-T. Hwang, S. M. Kang, J. E. Kay, T. Mauritsen, T. Ogura, G. Tselioudis, M. Watanabe, M. J. Webb, and A. A. Wing, 2026: The Cloud Feedback Model Intercomparison Project (CFMIP) contribution to CMIP7, *Geosci. Model Dev.*, submitted.

Thackeray, C., J. Norris, A. Hall, **M. D. Zelinka**, and S. Po-Chedley, 2026: Record setting heavy precipitation occurrence in 2024, *Sci. Adv.*, submitted.

Peer-Reviewed Publications

Asterisks indicate current or former advisees

110. Dingley, B., et al. including **M. D. Zelinka**, 2026: CMIP7 Data Request: Atmosphere Priorities and Opportunities, *Geosci. Model Dev.*, 19, 2945–2984, doi:10.5194/gmd-19-2945-2026.

109. **Zelinka, M. D.**, T. A. Myers*, Y. Qin*, L.-W. Chao*, S. A. Klein, S. Po-Chedley, P.-L. Ma, C. Wall, P. Ceppi, A. Gettelman, 2026: Recent cloud trends and extremes reaffirm established bounds on cloud feedback and aerosol-cloud interactions, *Commun. Earth Environ.*, doi:10.1038/s43247-026-03461-8.

108. Tam, R. Y. S., T. A. Myers*, **M. D. Zelinka**, C. Proistosescu, Y.-J. Lin, K. Marvel, 2026: Meteorological Drivers of the Low-Cloud Radiative Feedback Pattern Effect and its Uncertainty, *Atmos. Chem. Phys.*, 26, 4289–4311, doi:10.5194/acp-26-4289-2026.

107. Chao, L.-W.*, **M. D. Zelinka**, C. R. Terai, H. Beydoun, B. R. Hillman, N. D. Keen, P. M. Caldwell, P. A. Bogenschutz, and S. A. Klein, 2026: Examining Cloud Feedback Components in the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM), *J. Climate*, 39, 2563–2578, doi: 10.1175/JCLI-D-25-0656.1.

106. Zhang, Y., et al. including **M. D. Zelinka**, 2026: Understanding Biases in Simulated Cloud Radiative Effects in E3SMv3, *Earth Space Sci.*, doi:10.1029/2025EA004757.

105. Feng, C., et al. including **M. D. Zelinka**, 2026: Interconnection of Aerosol Cloud Interactions and Cloud Feedback through Warm Rain Process, *J. Geophys. Res.*, 131, doi:10.1029/2025JD044854.

104. Ceppi, P., S. Wilson Kemsley, H. Andersen, T. Andrews, R. J. Kramer, P. Nowack, C. J. Wall, and **M. D. Zelinka**, 2026: Emerging low-cloud feedback and adjustment in global satellite observations, *Atmos. Chem. Phys.*, 26, 4153–4171, doi:10.5194/acp-26-4153-2026.

103. Espinosa, Z.*, **M. D. Zelinka**, C. M. Bitz, K. C. Armour, 2026: The Strength of Coupling to the Southern Ocean Modulates Tropical Eastern Pacific Variability and Forced Response, *J. Climate*, 39, 1543–1558, doi:10.1175/JCLI-D-25-0226.1.

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101. Fridlind, A. M., et al. including **M. D. Zelinka**, 2025: Towards a Climate OSSE Framework for Satellite Mission Design, *Bull. Amer. Meteor. Soc.*, 107, E431–E445, doi:10.1175/BAMS-D-24-0242.1.

100. Aerenson, T. et al. including **M. D. Zelinka**, 2025: Global models predict clouds at the wrong time of day: does it matter for radiation and climate? *Sci. Adv.*, 11, eady3236, doi:10.1126/sciadv.ady3236.

99. Zehrung, A., A. D King, Z. Nicholls, **M. D Zelinka**, and M. Meinshausen, 2025: Standardising the "Gregory method" for calculating equilibrium climate sensitivity, *Geosci. Model Dev.*, 18, 9433–9450, doi:10.5194/gmd-18-9433-2025.

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97. Hill, P. G., D. L. Finney, and **M. D. Zelinka**, 2025: Cloud feedback uncertainty in the equatorial Pacific across CMIP6 models, *Geophys. Res. Lett.*, 52, doi:10.1029/2025GL117183.

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93. Zhou, C.*, Q. Wang, I. Tan, L. Zhang, **M. D. Zelinka**, M. Wang, and J. Bloch-Johnson, 2025: Sea ice pattern effect on Earth's energy budget is characterized by hemispheric asymmetry, *Sci. Adv.*, 11, doi:10.1126/sciadv.adr4248.

92. **Zelinka, M. D.**, L.-W. Chao*, T. A. Myers*, Y. Qin*, and S. A. Klein, 2025: Technical note: Recommendations for diagnosing cloud feedbacks and rapid cloud adjustments using cloud radiative kernels, *Atmos. Chem. Phys.*, 25, 1477-1495, doi:10.5194/acp-25-1477-2025.

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4. **Zelinka, M.D.** and D.L. Hartmann, 2012: Climate Feedbacks and their Implications for Poleward Energy Flux Changes in a Warming Climate. *J. Climate*, 25, 608–624, doi:10.1175/JCLI-D-11-00096.1.

3. **Zelinka, M.D.** and D.L. Hartmann, 2011: The Observed Sensitivity of High Clouds to Mean Surface Temperature Anomalies in the Tropics. *J. Geophys. Res.*, 116, D23103, doi:10.1029/2011JD016459.

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Book Chapters

Chao, L. -W.*, A. E. Dessler, and **M. D. Zelinka**, 2026: Climate Feedbacks. In: Robinson, W.A. (Ed.), *Encyclopedia of Atmospheric Sciences*, vol. 5. Elsevier, Academic Press, pp. 117–128, doi:10.1016/B978-0-323-96026-7.00038-2.

McCoy, D. T., M. E. Frazer, J. Muelmenstaedt, I. Tan, C. R. Terai, and **M. D. Zelinka**, 2024: Extratropical Cloud Feedbacks, in *Clouds and their Climatic Impacts: Radiation, Circulation, and Precipitation*, S. C. Sullivan (Ed) and C. Hoose (Ed), American Geophysical Union.

Contributing author to Forster et al. 2021: The Earth's Energy Budget, Climate Feedbacks, and Climate Sensitivity, in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.* Masson-Delmotte et al. (eds.). Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, pp. 923-1054, doi:10.1017/9781009157896.009.

McCoy, D.T., D. L. Hartmann, and **M. D. Zelinka**, 2017: Mixed-Phase Cloud Feedbacks, in *Mixed-phase Clouds: Observations and Modeling*, Andronache, C. (Ed.), Elsevier.

Tan, I., T. Storelvmo, and **M. D. Zelinka**, 2017: The climatic impact of thermodynamic phase partitioning in mixed-phase clouds, in *Mixed-phase Clouds: Observations and Modeling*, Andronache, C. (Ed.), Elsevier.

Dessler, A.E. and **M. D. Zelinka**, 2015: Climate Feedbacks, in *Encyclopedia of Atmospheric Sciences, 2nd edition*, Vol 2, pp. 18-25, G. R. North (editor-in-chief), J. Pyle and F. Zhang (editors).

Recent Honors & Awards

Richard P. and Linda S. Turco Lectureship, Department of Climate, Meteorology, & Atmospheric Sciences, University of Illinois Urbana–Champaign, 2024-2025.

LLNL Physical and Life Sciences Directorate Awards for Excellence in Publications for Lee et al. (2024), Sherwood et al. (2020), Zhou et al. (2016), and Norris et al. (2016)

Zelinka et al. (2020) named "Editor's Choice Paper" by Editor-in-Chief of *Geophysical Research Letters*, 2024

American Meteorological Society Henry G. Houghton Award, 2022

Eos Research Spotlights for Zelinka et al. (2022), Zelinka et al. (2020), Zhou et al. (2017), Zelinka et al. (2016), and McCoy et al. (2016)

Editors' Citation for Excellence in Refereeing - *Geophysical Research Letters*, 2021

LLNL Deputy Director's Science & Technology Excellence in Publication Awards for Zelinka et al. (2020), Sherwood et al. (2020), and Santer et al. (2015)

Sherwood et al. (2020) named runner-up for *Science Magazine*'s 2020 Breakthrough of the Year

Nature Climate Change Research Highlight for Dong et al. (2020)

US CLIVAR Research Highlight for Zelinka et al. (2020)

LLNL Outstanding Mentor Award, 2018

LLNL Early and Mid-Career Recognition Program Award, 2018

Diablo Magazine 40 Under 40, 2017

Professional Activities, Service, & Leadership Roles

Chair, 2027 Gordon Research Conference on Radiation and Climate

Vice Chair, 2025 Gordon Research Conference on Radiation and Climate

Editor, *Journal of Climate*, Dec 2024–present

Steering Group, CMIP7 Data Request (Atmosphere Theme), Jul 2024–present

External Peer Reviewer, NASA Langley Research Center Science Directorate, Nov 2023

Programme Advisory Group, Uncertainty in Climate Sensitivity due to Clouds (CloudSense) Research Programme, Sep 2023–present

CFMIP Scientific Steering Committee, Jul 2023–present

Discussion Leader, 2023 Gordon Research Conference on Radiation & Climate

AGU Global Environmental Change Fellows Committee, 2022–2024

Contributing Author, IPCC 6th Assessment Report, 2019–2021

LLNL Physical and Life Sciences Postdoc Committee, 2017–2020

Section Editor, *Current Climate Change Reports* Topical Collection on Climate Feedbacks, 2014–2017

Contributor, climatefeedback.org, 2017–present

Chair, 2011 Gordon Research Seminar on Radiation and Climate

Proposal reviewer for DOE, European Research Council, NASA, and NSF

Editor, *Proceedings of the National Academy of Sciences*

Reviewer for:
Atmosphere | Atmos. Ocean | Atmos. Chem. Phys. | Atmos. Meas. Tech. | Atmos. Sci. Lett.
B Am Meteorol Soc | Clim. Dynam. | Climatic Change | Earth's Future | Earth Syst. Dyn.
Earth Syst. Sci. Data | Environ. Res. Lett. | Geophys. Res. Lett. | Geosci. Model Dev.
Int. J. Appl. Earth Obs. Geoinf. | J. Adv. Model. Earth Syst. | J. Appl. Meteorol. Clim.
J. Atmos. Oceanic Technol. | J. Atmos. Sci. | J. Climate | J. Geophys. Res | J. Meteorol. Soc. Jpn.
Nature | Nat. Clim. Change| Nat. Commun. | Nat. Geosci. | npj Climate and Atmospheric Science
P. Natl. Acad. Sci. | Sci. Rep. | Sci. Adv. | Surv. Geophys.

Recent Invited Presentations

UC San Diego, Scripps Institution of Oceanography, 27 Jan 2026

McGill University, Department of Atmospheric and Oceanic Sciences, 21 Oct 2025

Duke University, Earth & Climate Sciences Division, 3 Oct 2025

Stanford University, Department of Earth System Science, 2 Apr 2025

Univ. of Illinois Urbana–Champaign, Dept. of Climate, Meteorology, & Atmospheric Sciences, 22 Oct 2024

CERES Science Team Meeting, 2 Oct 2024

LLNL Senior Staff Meeting, 1 Jul 2024

NASA Ames Research Center, Earth Science Division, 20 Jun 2024

Yale University, School of the Environment, 25 Apr 2024

AGU Fall Meeting: The Flows of Energy Through the Climate System Session, 12 Dec 2023

AGU Fall Meeting: Climate Sensitivity and Feedbacks Session, 11 Dec 2023

University of Cambridge Centre for Atmospheric Science, Department of Chemistry, 24 Oct 2023

Yale University School of the Environment, 27 Apr 2023

AGU Fall Meeting: Atmospheric Physics, Radiation, Clouds, and Aerosols Session, 15 Dec 2022

Yale University, School of the Environment, 21 Apr 2022

Aerosol and Cloud, Convection and Precipitation Webinar Series, 19 Apr 2021

University of Maryland Baltimore County Department of Physics Colloquium, 24 Feb 2021

University of Toronto Physics Colloquium, 28 Jan 2021

Advisees

Postdocs

Li-Wei Chao (Postdoc, LLNL)

Yi Qin (Research Scientist, Laoshan Laboratory)

Timothy Myers (Research Scientist, NOAA/CIRES)

Chen Zhou (Associate Professor, Nanjing University)

Graduate Students

Zac Espinosa (University of Washington)

Undergraduate Students

Russell Hunter (Duke Univesity / Second Lieutenant, US Space Force)

Thea Moellerstedt (Graduate Student, UCSD)

Scott Feldman (Meteorologist, Verisk Weather Solutions)

Outreach Activities

LLNL Ambassador Lecturer, Aug 2024 - present

LLNL Technical Lightning Talk: Community Cohorts program for new hires, 24 Aug 2023.

LLNL - Las Positas College Science and Engineering Seminar Series, 20 Apr 2023.

Science on Saturday, The Future in Focus: Predicting Climate Change through Observations, Modeling, and Artificial Intelligence, 4 Mar 2023.

Castro Valley Rotary Club, 13 Sep 2022.

Science on Saturday, The Future in Focus: Predicting Climate Change through Observations, Modeling, and Artificial Intelligence, 26 Feb 2022.

San Joaquin County Office of Education Climate Change Summit, 26 Sep 2020

Panel Discussant, Wild and Scenic Film Festival, Bankhead Theater, Livermore, CA, Jan 2020

Univ. of Washington Dept. of Atmospheric Sciences and Program on Climate Change Outreach Teams, Sep 2004 – Dec 2010

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